

**Joint modeling of the upper and lower tail dependence in
daily log-return of Indigo and Spicejet airlines during
COVID-19**

MTH697A Course Project

of

BS-MS Dual Degree Physics

by

**Md Kaif Faiyaz
Roll No: 19817491**

Under the guidance of

Professor Arnab Hazra



**DEPARTMENT OF MATHEMATICS & STATISTICS
INDIAN INSTITUTE OF TECHNOLOGY KANPUR**

DECLARATION

I hereby declare that the work presented in the project report entitled **Joint modeling of the upper and lower tail dependence in daily log-return of Indigo and Spicejet airlines during COVID-19** contains my own ideas in my own words. At places, where ideas and words are borrowed from other sources, proper references, as applicable, have been cited. To the best of my knowledge, this work does not emanate from or resemble other work created by person(s) other than mentioned herein.

Name: **Md. Kaif Faiyaz**

Date: November 3, 2023

Name: **Arnab Hazra (Supervisor)**

Date: November 3, 2023

Contents

Introduction	1
Copula Distribution	2
GEV distribution	3
Stationarity of Data	4
Data Preprocessing	7
Mathematical Modelling	9
Inference	13
Conclusion	16
Acknowledgements	17
Bibliography	18

ABSTRACT

The year 2020 witnessed the rapid global spread of the COVID-19 virus, compelling governments around the world to impose a series of stringent measures, including the suspension of both international and domestic flights, in an effort to contain the virus's transmission. The volatile nature of the pandemic, characterized by intermittent easing and reinstatement of travel restrictions, sent shockwaves through the global stock markets.

This research project takes a focused approach, centering on the joint volatility observed in the stock prices of two major Indian private airlines, Indigo and SpiceJet, with respective market shares of 48.2% and 15.6%. While existing literature often emphasizes the dynamic mean processes in such datasets, this study takes a unique perspective by analyzing tail-dependence, crucial for understanding the stochastic behavior in both the lower and upper tails of stock prices. This analysis is particularly pertinent since buy positions are sensitive to falling prices (lower tail), while sell positions are impacted by price increases (upper tail).

To assess these different types of risk comprehensively, the study employs a recently developed copula model capable of capturing both lower and upper-tail asymptotic dependence or independence in a unified manner. The daily log-return data on average prices for Indigo and SpiceJet Airlines serves as the basis for this model application, and simulation studies were conducted to validate its efficacy before being employed on real data.

This research's practical significance lies in its ability to detect market risk contagion, making it a valuable source of information for investors seeking to diversify their portfolios in an ever-evolving economic landscape. By shedding light on the tail behavior of these airlines' stock prices, this study contributes to more informed investment strategies in times of economic volatility and uncertainty.

INTRODUCTION

The year 2020 bore witness to the unprecedented COVID-19 pandemic, a global crisis triggered by the emergence of the novel coronavirus, SARS-CoV-2. This report delves into the profound consequences of the pandemic on its devastating effects on the aviation sector. We introduce a versatile dependence model to examine the tail behavior of stock prices for 2 of India's major airlines: Indigo and SpiceJet Airlines. **WHO** officially designate the outbreak as a pandemic on March 11, 2020. To curb the virus's transmission, many countries implemented extensive measures, including partial or complete lockdowns and travel restrictions.

The economic repercussions of the pandemic were significant, as it disrupted global supply chains, tourism, and international trade. The aviation sector, in particular, faced substantial challenges due to its inherent vulnerability to external factors. The cessation of tourism and international travel, coupled with the heightened risk profile for airline companies, had far-reaching consequences on the industry.

We delve into a single flexible dependence model capable of examining the tail behavior of financial data. It offers high flexibility in assessing both the lower and upper tails of data distributions, facilitating a transition between the 2 extremes.

To demonstrate the model's applicability, we used it to analyze the stock price **return rate** data of the daily maximum and minimum price for two prominent Indian airlines: Indigo and SpiceJet. By applying the model, we were able to infer the tail behavior of these airline stocks. This analysis can provide valuable insights for investors and stakeholders in these companies.

The report presents the statistical findings and model outcomes for the two airlines, highlighting the observed tail behavior. The discussion section delves into the implications of these findings for investors, emphasizing the importance of understanding tail behavior when making investment decisions.

COPULA DISTRIBUTION

Copula is a powerful way to model the dependence of a random vector. One key insight is due to the famous **Sklar theorem**: the distribution of any continuous random vector can be expressed using copula and the marginal distribution.

For a random vector $\mathbf{X}$ with CDF F and univariate marginal CDFs $F_1, \dots, F_d, \exists$, a copula C such that

$$F(x_1, \dots, x_d) = C(F_1(x_1), \dots, F_d(x_d)).$$

If X is continuous, then such a copula C is unique. This shows that we can write the joint CDF in terms of marginal CDFs and the copula. Conversely, if we know the joint CDF F and the marginals $F_1, \dots, F_d$, we can find the copula via

$$C(u_1, \dots, u_d) = F(F_1^{-1}(u_1), \dots, F_d^{-1}(u_d))$$

Given that a copula can be viewed as a CDF of a uniform random vector, we can then think about the corresponding PDF. Such density function is known as a **copula density**. The Sklar theorem implies that the PDF of X can be written as

$$p(x_1, \dots, x_d) = c(F_1(x_1), \dots, F_d(x_d))p_1(x_1) \cdots p_d(x_d)$$

, where $p_j(x_j)$ is the PDF of X_j and c is the copula density.

Some famous families of copulas are: (i) Archimedean copula, (ii) Bi-variate Gaussian copula (iii) EFGM (Eyrraud-Farlie-Gumbel-Morgenstern) copula.

GENERALIZED EXTREME VALUE DISTRIBUTION

Generalized extreme value (**GEV**) distribution is a family of continuous probability distributions developed within extreme value theory to combine the **Gumbel**, **Fréchet** & **Weibull** also known as type I, II & III extreme value distributions.

The Extreme Value Theory (EVT) deals with extreme observations; they are usually defined as block maxima (e.g., annual maxima of daily observations) or threshold exceedances. The important aspect of EVT lies in describing the tail behavior of a stochastic process. Mathematically, consider $X_1, X_2, \dots$, a sequence of independent random variables, where X_i 's are measured on a regular time scale, following a common continuous CDF $F_0()$.

The main goal of EVT is to study the asymptotic behavior of $M_n = \max_n \{X_1, X_2, \dots, X_n\}$. The CDF of M_n is:

$$P(M_n \leq z) = F_0^n(z)$$

In practice, $F_0()$ is generally unknown. In this context, the celebrated **Fisher-Tippett** theorem states that for any arbitrary $F_0()$, if there exist sequences of real numbers $\{a_n^0\}$ and $\{b_n^0\}$ and a non-degenerate CDF $G_0()$ satisfying $\lim_{n \rightarrow \infty} F_n^0(a_n^0 x + b_n^0) = G_0(x)$ pointwise, then $G_0()$ must belong to either Gumbel, Fréchet, or Weibull families. The GEV distribution is finally given by:

$$G_0(y) = \exp \left[- \left\{ 1 + \xi \left(\frac{y - \mu}{\sigma} \right) \right\}_+^{\frac{-1}{\xi}} \right]$$

here $\mu \in \mathbb{R}$, $\sigma \in \mathbb{R}^+$, and $\xi \in \mathbb{R}$ are the location, scale, and shape parameters of the GEV distribution, and $x_+ = \max(x, 0)$. Depending on whether the shape parameter is zero, positive, or negative, the GEV family is called the Gumbel, Fréchet, or Weibull family, respectively. We denote the distribution with $GEV(\mu, \sigma, \xi)$, and in practice, we assume $M_n \tilde{GEV}(\mu, \sigma, \xi)$ in a limiting sense. The centering and scaling of M_n in the Fisher-Tippett theorem are assumed to be adjusted by the parameters μ and σ , respectively.

In R programming language the **evgam** package is used for GEV distribution.

STATIONARITY OF DATASET

A stationary process is a stochastic process whose unconditional joint probability distribution does not change when shifted in time. It has the following properties:

1. Constant mean 2. Constant variance 3. Doesn't have seasonality, i.e., does not depend on the time at which the series is observed. In general, a stationary time series will have no predictable patterns in the long-term. The statistical test employed to test the stationarity of the stock price data from 2019-12-02 to 2022-05-31 **Augmented Dickey-Fuller** test is employed.

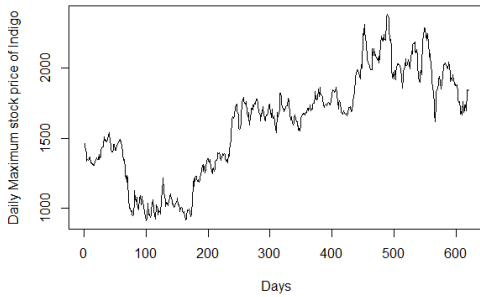
Stock Price	ADF p-value
Indigo daily maximum stock price	0.3931
Indigo daily minimum stock price	0.2974
SpiceJet daily maximum stock price	0.4511
SpiceJet daily minimum stock price	0.3965

Table 1: ADF Test statistic for daily maximum and minimum for Indigo & SpiceJet

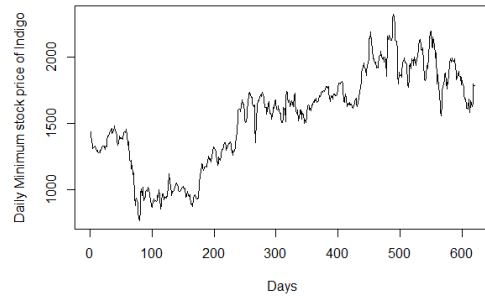
ADF test is conducted with the following hypotheses:

1. **Null**: Series is non-stationary. 2. **Alternate**: Series is stationary

As the p-value for minimum & maximum of the stock price data for Indigo & SpiceJet is ≥ 0.05 the null hypothesis is failed to be rejected. Hence they are non-stationary data.



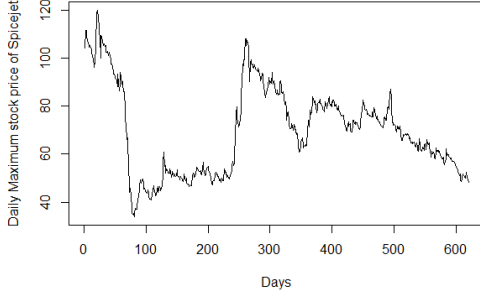
(a) Daily maximum stock price of Indigo



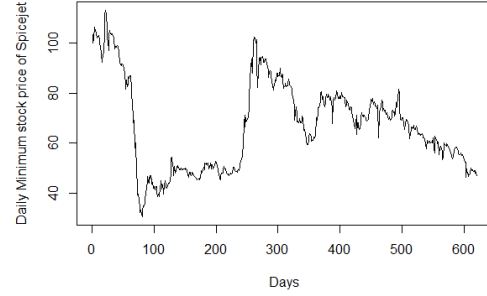
(b) Daily minimum stock price of Indigo

Figure 1: Daily Stock prices of Indigo airlines from 2019-12-02 to 2022-05-31

Hence we work on analyzing the return rates of stock prices having more meaningful comparison metrics.



(a) Daily maximum stock price of SpiceJet



(b) Daily minimum stock price of Spicejet

Figure 2: Daily Stock prices of SpiceJet airlines from 2019-12-02 to 2022-05-31

Definition 0.1. The **return rate** is the percentage variation in the value of a particular stock investment over a specified period, indicating the level of gain or loss experienced by the investor on their stock holding. Mathematically:

$$RR = \frac{E - B + D}{B} \quad (1)$$

where **RR** is the return rate, **E** is the ending price of the period, **B** is the beginning price of the period & **D** is total amount of dividends paid out by the company during that same period. The daily maximum return rate is defined as follows:

Suppose $Y_t(s)$: stock price on a specific day **t** at time **s**.

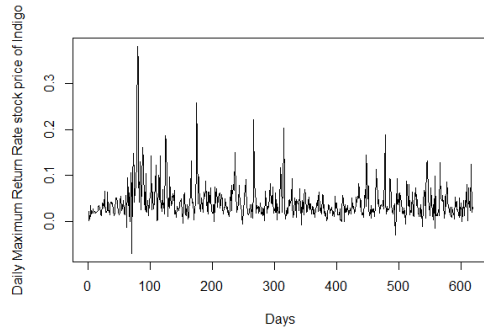
We calculate the daily maximum stock return rate $R_t^{\max}$ on day **t** as:

$$R_t^{\max} = \max_{s,s'} \frac{Y_{t+1}(s) - Y_t(s')}{Y_t(s')} = \frac{\max_s Y_{t+1}(s)}{\min_{s'} Y_t(s')} - 1 \quad (2)$$

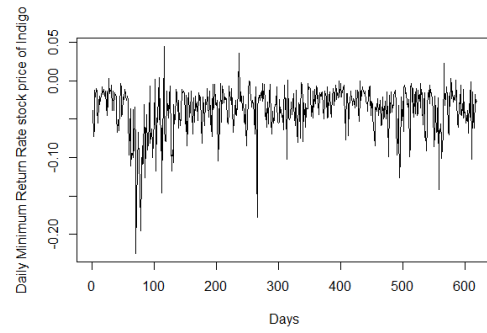
Similarly the daily minimum stock return rate $R_t^{\min}$ on day **t** as:

$$R_t^{\min} = \min_{s,s'} \frac{Y_{t+1}(s) - Y_t(s')}{Y_t(s')} = \frac{\min_s Y_{t+1}(s)}{\max_{s'} Y_t(s')} - 1 \quad (3)$$

As the data for daily maxima and minima of stock prices is publicly available the above calculations are easy to do for each of the airlines. The daily maximum return rate profiles are relatively more stable than the actual stock prices. However, a more jittery pattern is observable for SpiceJet than that for Indigo. During the initial days of the COVID-19 pandemic in India, R_t profiles attend negative values for both airlines; however, the lowest value of R_t is more negative for SpiceJet than that for Indigo. A clear nonstationary pattern is observable in both series

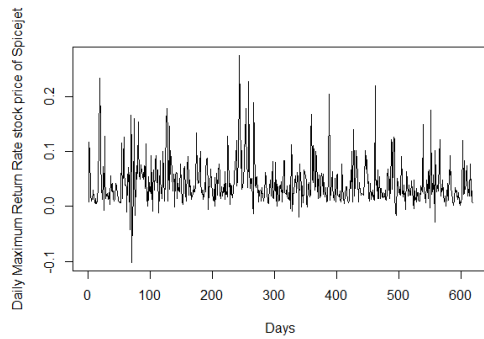


(a) Daily maximum stock return rate

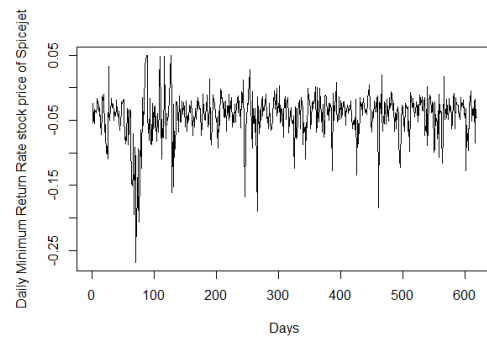


(b) Daily minimum stock return rate

Figure 3: Daily Stock Return rate of Indigo airlines from 2019-12-02 to 2022-05-31



(a) Daily maximum stock return rate



(b) Daily minimum stock return rate

Figure 4: Daily Stock Return rate of SpiceJet airlines from 2019-12-02 to 2022-05-31

DATA PREPROCESSING

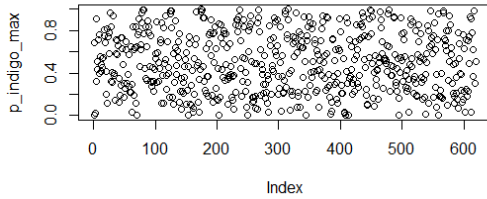
We try to convert the Indigo and SpiceJet maximum and minimum return rates to $U(0,1)$ by fitting the **Generalized Extreme Value** distribution. For each of the airlines and the maximum as well as the minimum return rate for the airlines we try to fit the splines on the index and try to find the optimal **kernel** value using **bootstrap**.

For the bootstrap we divide the dataset into training and test set in the ration of 9 : 1. In other words we divide the dataset into intervals of 10 and for the i^{th} bootstrap we select the i^{th} element from each interval of 10 elements. Then we calculate the test statistics and find the most appropriate value of **k** such that maximum and minimum stock return rates for Indigo & SpiceJet are as **Uniform** as possible.

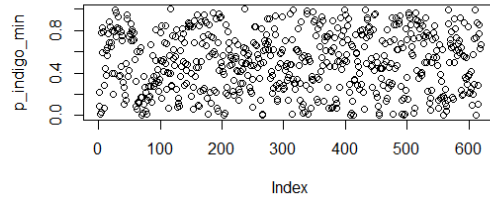
Return Rate Stock Price	k value for spline
Indigo maximum stock return rate	15
Indigo minimum return rate	15
SpiceJet maximum return rate	24
SpiceJet minimum return rate	29

Table 2: Kernel value for stock Return Rate in the formula of **GEV** distribution

We concatenate these transformed datasets so as to carryout joint modelling the upper and lower tails of the Indigo & SpiceJet airlines. The histogram & scatter plots for the transformed return rates dataset is:

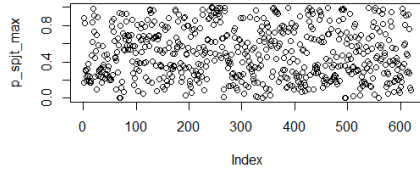


(a) Scatter plot of transformed maximum stock return rate

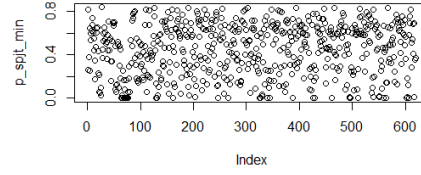


(b) Scatter plot of transformed minimum stock return rate

Figure 5: Transformed Stock Return rate of Indigo airlines from 2019-12-02 to 2022-05-31

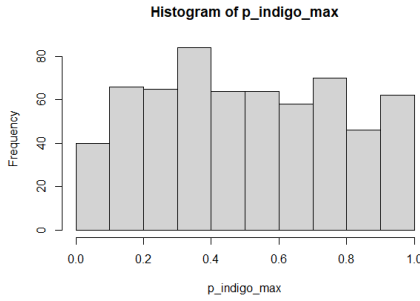


(a) Scatter plot of transformed maximum stock return rate

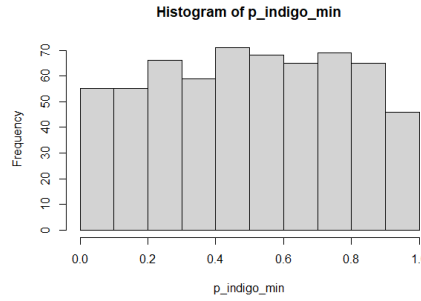


(b) Scatter plot of transformed minimum stock return rate

Figure 6: Transformed Stock Return rate of SpiceJet airlines from 2019-12-02 to 2022-05-31

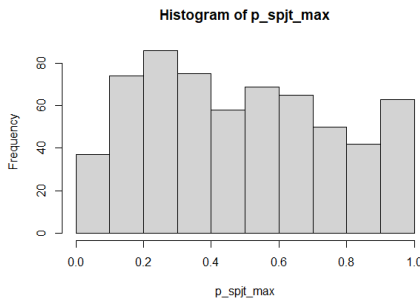


(a) Histogram of transformed maximum stock return rate

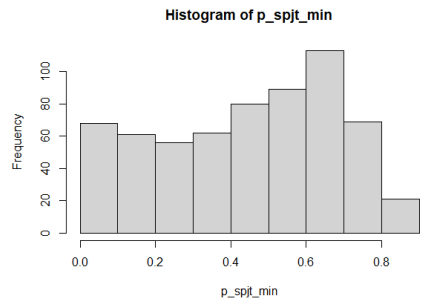


(b) Histogram of transformed minimum stock return rate

Figure 7: Transformed Stock Return rate of Indigo airlines from 2019-12-02 to 2022-05-31



(a) Histogram of transformed maximum stock return rate



(b) Histogram of transformed minimum stock return rate

Figure 8: Transformed Stock Return rate of SpiceJet airlines from 2019-12-02 to 2022-05-31

MATHEMATICAL MODELLING

We empirically check the **upper-upper** & **lower-lower** tail dependence coefficients using the following R code:

$$\frac{\text{mean}((p_{\text{indigo-max}} > 0.95) \text{ and } (p_{\text{spjt-max}} > 0.95))}{0.05} ; \frac{\text{mean}((p_{\text{indigo-min}} < 0.05) \text{ and } (p_{\text{spjt-min}} < 0.05))}{0.05}$$

and get the following results of tail dependence:

Lower-Lower: 0.16155089; **Upper-Upper** :0.3231018

where $p_{\text{indigo-max}}$ is the uniform transformed return rate of daily maximum stock price for Indigo & $p_{\text{indigo-min}}$ is the uniform transformed return rate of daily minimum stock price for Indigo. Similarly the other 2 terms are for SpiceJet dataset.

From the above results obtained empirically there is dependence in return rate of both airlines in their upper & lower tails. Now we will see a flexible **COPULA MODEL** which is able to distinctively capture asymptotic dependence or independence in its lower and upper tails simultaneously. This model is parsimonious and smoothly bridges (in each tail) both extreme dependence classes in the interior of the parameter space.

Model Construction

Let R be random variable with **Asymmetric Laplace** distribution, denoted by F_R given by $AL(\delta_l, \delta_u)$,

$$F_R(r) = \begin{cases} \delta_l \frac{e^{\frac{r}{\delta_l}}}{\delta_l + \delta_u} & r \leq 0 \\ 1 - \frac{\delta_u e^{-\frac{r}{\delta_u}}}{\delta_l + \delta_u} & r > 0 \end{cases}$$

where $\delta_l, \delta_u \in (0, 1)$ are scale parameters of lower and upper tails, respectively. Let W_1, W_2 have $AL(1 - \delta_l, 1 - \delta_u)$ distribution, then we assume that the random vector $\mathbf{W} = [W_1 \ W_2]^T$ follows **Gaussian Copula model** with correlation $\rho \in (-1, 1)$. The joint distribution of $\mathbf{W}$ satisfies:

$$P(W_1 \leq w_1, W_2 \leq w_2) = \Phi_\rho [\Phi^{-1}\{F_W(w_1)\}, \Phi^{-1}\{F_W(w_2)\}]$$

where Φ and Φ_ρ denote the uni-variate standard Gaussian distribution and bi-variate standard Gaussian distribution with correlation ρ , respectively. Our model through the random vector $\mathbf{X} = [X_1 \ X_2]^T$ where $X_i = R + W_i$. The dependence structure of $\mathbf{X}$ interpolates between that $\mathbf{R}$ (perfect dependence) and that of $\mathbf{W}$ (independence).

Hence our model is constructed on the basis of three-parameters $(\delta_l, \delta_u, \rho)$ which helps us to evaluate the dependence structure in both lower and upper tail.

Note: Any other model, other than **Gaussian copula** model could be used that is asymptotically independent in both tails. This model has been used for computational convenience.

Expressions associated with Copula Model

The Copula model we employ was used by Gong & Heuser in their research paper "Asymmetric tail dependence modeling, with application to cryptocurrency market data". Now we state all the expressions of marginals and joint distributions deduced using above model and used in our source code. The pdf of the Asymptotic Laplace Distribution is $f_R(r)$ is given as:

$$f_R(r) = \begin{cases} \frac{e^{\frac{r}{\delta_l}}}{\delta_l + \delta_u} r \leq 0 \\ \frac{e^{\frac{-r}{\delta_u}}}{\delta_l + \delta_u} r > 0 \end{cases}$$

We evaluate the marginals for X_i under various scenarios of δ_l, δ_u

$$F_X(x) = P(X_i \leq x) = P(R + W_i \leq x) = \int_{\mathbb{R}} P(W_i \leq (x - r)) f_R(r) dr$$

Case 1: $\delta_l \neq \frac{1}{2}, \delta_u \neq \frac{1}{2}$

$$F_X(x) = \begin{cases} K_1 e^{\frac{x}{\delta_l}} - K_2 e^{\frac{x}{1-\delta_l}} & x \leq 0 \\ 1 + K_3 e^{\frac{-x}{\delta_u}} - K_4 e^{\frac{-x}{1-\delta_u}} & x > 0 \end{cases}$$

$$\begin{aligned} K_1 &= \frac{\delta_l^3}{(\delta_l + \delta_u)(2\delta_l - 1)(1 + \delta_l - \delta_u)}; K_2 = \frac{(\delta_l - 1)^3}{(\delta_l - \delta_u - 1)(2\delta_l - 1)(2 - \delta_l - \delta_u)} \\ K_3 &= \frac{\delta_u^3}{(\delta_l + \delta_u)(2\delta_l - 1)(\delta_l - \delta_u - 1)}; K_4 = \frac{(\delta_u - 1)^3}{(1 + \delta_l - \delta_u)(2\delta_l - 1)(2 - \delta_l - \delta_u)} \end{aligned}$$

Case 2: $\delta_l = \frac{1}{2}, \delta_u \neq \frac{1}{2}$

$$F_X(x) = \begin{cases} K_5 x e^{2x} - K_6 e^{2x} & x \leq 0 \\ 1 - K_7 e^{\frac{x}{\delta_u}} - K_8 e^{\frac{-x}{1-\delta_u}} & x > 0 \end{cases}$$

$$\begin{aligned}
K_5 &= \frac{2}{(1+2\delta_u)(2\delta_l-3)}; K_6 = \frac{(-12\delta_u + 12\delta_u^2 - 5)}{(1+2\delta_u)^2(2\delta_l-3)^2} \\
K_7 &= \frac{4\delta_l^3}{(1+2\delta_u)^2(2\delta_u-1)}; K_8 = \frac{4(\delta_u-1)^3}{(2\delta_u-3)^2(2\delta_u-1)}
\end{aligned}$$

Case 3: $\delta_l \neq \frac{1}{2}, \delta_u = \frac{1}{2}$

$$F_X(x) = \begin{cases} K_9 x e^{\frac{x}{\delta_l}} + K_{10} e^{\frac{x}{1-\delta_l}} & x \leq 0 \\ 1 + K_{11} x e^{-2x} + K_{12} e^{-2x} & x > 0 \end{cases}$$

$$\begin{aligned}
K_9 &= \frac{4\delta_l^3}{(1+2\delta_l)^2(2\delta_l-1)}; K_{10} = \frac{4(\delta_u-1)^3}{(2\delta_l-3)^2(2\delta_l-1)} \\
K_{11} &= \frac{2}{(1+2\delta_l)^2(2\delta_u-3)}; K_{12} = \frac{4(\delta_u-1)^3}{(2\delta_l-3)^2(2\delta_l+1)}
\end{aligned}$$

Case 4: $\delta_l = \frac{1}{2}, \delta_u = \frac{1}{2}$

$$F_X(x) = \begin{cases} \frac{1}{2}(1-x)e^{2x} & x \leq 0 \\ 1 - \frac{1}{2}(1+x)e^{-2x} & x > 0 \end{cases}$$

The density function f_X is easily obtained by differentiating F_X . The joint distribution of $F_X(x_1, x_2)$ is:

$$F_X(x_1, x_2) = P(X_1 \leq x_1, X_2 \leq x_2) = \int_{\mathbb{R}} \Phi_{\rho} [\Phi^{-1}\{F_W(x_1-r)\}, \Phi^{-1}\{F_W(x_2-r)\}] f_R(r) dr$$

. We employ numerical integration for implementation of quantile distribution function of Gaussian distribution. Now to get joint density function we need to differentiate above expression with respect to x_1 and x_2 .

$$f_X(x_1, x_2) = \int_{\mathbb{R}} \Phi_{\rho} [\Phi^{-1}\{F_W(x_1-r)\}, \Phi^{-1}\{F_W(x_2-r)\}] \Pi_{i=1}^2 \frac{f_W(x_i-r)}{\phi\{F_W(x_i-r)\}} f_R(r) dr$$

Tail dependence structures

We explore the lower- and upper-tail properties of the proposed model. We consider, for each threshold $t \in (0, 1)$, the tail coefficients:

$$\begin{aligned}
\chi_l(t) &= P(U_1 < t | U_2 < t) = \frac{C(t,t)}{t} \\
\chi_u(t) &= P(U_1 > t | U_2 > t) = \frac{1-2t+C(t,t)}{1-t}
\end{aligned}$$

, where $\chi_l(t), \chi_u(t)$ are the lower and upper tail coefficients & $U_i = F_{X_i}$. The **Asymptotic dependence/Independence** class (AD/AI) can be determined by computing limits $\lim_{t \rightarrow 0} \chi_l(t) > 0$ and $\lim_{t \rightarrow 0} \chi_u(t) > 0$. We dig deeper into the dependence classes with the following proposition:

For our Copula model random vector $\mathbf{X}$ has following dependence cases:

Case(1): $\delta_{l/u} \leq 0.5$ Then, $\mathbf{X}$ is asymptotically independent in its lower/upper tail with $\chi_{L/U} = 0$ and coefficient of lower/upper tail dependence obtained as:

$$\eta_{l/u} = \begin{cases} \frac{\delta_{l/u}}{1 - \delta_{l/u}} & \delta_{l/u} > \frac{1 + \rho}{3 + \rho} \\ \frac{1 + \rho}{2} & \delta_{l/u} \leq \frac{1 + \rho}{3 + \rho} \end{cases}$$

Case(2): $\delta_{l/u} > 0.5$ Then, $\mathbf{X}$ is asymptotically dependent in its lower/upper tail with coefficient of lower/upper tail dependence $\eta_{l/u} = 1$ and writing $s_l = -1, s_u = 1$ we have

$$\chi_{l/u} = E \left(\min \left[\frac{e^{\frac{s_{l/u} W_1}{\delta_{l/u}}}}{E \left\{ e^{\frac{s_{l/u} W_1}{\delta_{l/u}}} \right\}}, \frac{e^{\frac{s_{l/u} W_2}{\delta_{l/u}}}}{E \left\{ e^{\frac{s_{l/u} W_2}{\delta_{l/u}}} \right\}} \right] \right)$$

INFERENCE

Parameter Estimation

Let $Y_1, \dots, Y_n$ be n instances of random vector of the form $\mathbf{Y} = [Y_1 \ Y_2]^T$ that has the same copula models described above but possesses potentially different marginal distributions. In other words, the joint distribution of $\mathbf{Y}$ may be expressed as

$$F_Y(y_1, y_2) = Pr(Y_1 \leq y_1, Y_2 \leq y_2) = C \{F_{Y,1}(y_1), F_{Y,2}(y_2)\}$$

where C is our copula model. In order to estimate the underlying copula C from an observed random sample we adopt a two-step estimation approach.

Initially we estimate marginal distributions and transform the data to the standard uniform scale. This is achieved as discussed in the section of Data Preprocessing.

Secondly, we adopt a local (parametric or nonparametric) estimation approach for a time varying copula structure due to nonstationary of data over time. To estimate the dependence parameters, we here address this issue by proposing a local copula-based likelihood estimation approach that can capture complex trends in a very flexible way. For each time point to do this, we replace the likelihood function by a family of weighted local likelihoods to be maximized, which have the form:

$$L(\Theta_i) = \prod_{j=1}^n \omega(|j - i|) L_j(\Theta_i)$$

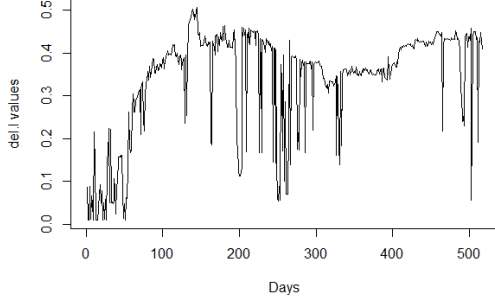
$\forall i \in 1, 2, \dots, n$, where $\omega(|j - i|) = 1$ if $|j - i| \leq 50$ & 0 otherwise. The L'_j s are defined:

$$L_j = c(u_{j,1}, u_{j,2}) = \frac{f_X \{F_X^{-1}(u_1), F_X^{-1}(u_2)\}}{f_X \{F_X^{-1}(u_1)\} f_X \{F_X^{-1}(u_2)\}}$$

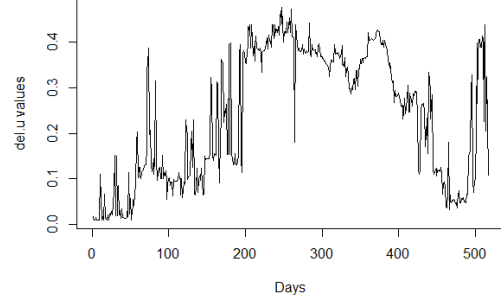
where 1, 2 represent the Indigo and SpiceJet airlines.

For each time stamp we use the likelihood as defined above by employing the upper-upper and lower-lower tail values for the airlines concurrently. Hence we get the following time-varying plot of the parameters δ_l, δ_u, ρ .

The plots obtained for the parameters are quite jittery possibly due to taking only about 100 data points for local likelihood estimation. Hence we smoothen out the curves. We use splines & GAM package in R to fit a smooth curve to the obtained plots of parameters.

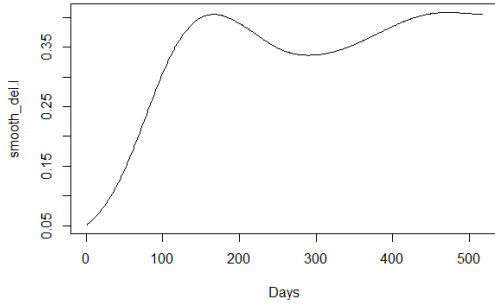


(a) δ_l vs days from 21-01-2020 to 10-04-2022

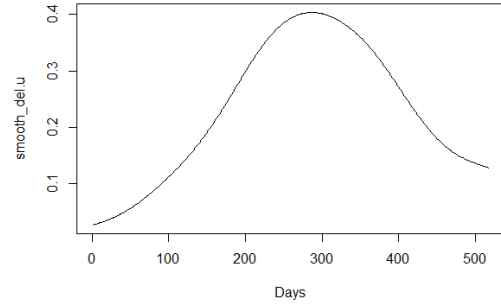


(b) δ_u vs days from 21-01-2020 to 10-04-2022

Figure 9: Time varying plots of parameters δ_l, δ_u



(a) δ_l (smooth curve) vs days from 21-01-2020 to 10-04-2022



(b) δ_u (smooth curve) vs days from 21-01-2020 to 10-04-2022

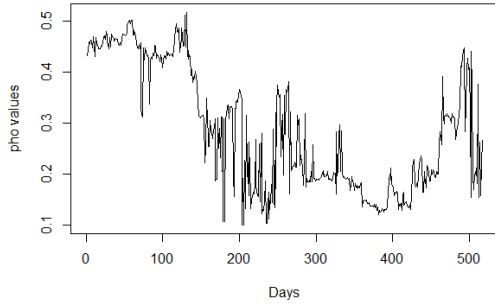
Figure 10: Time varying plots of parameters δ_l, δ_u (smooth curve)

Tail Dependence structure and Tail Coefficient

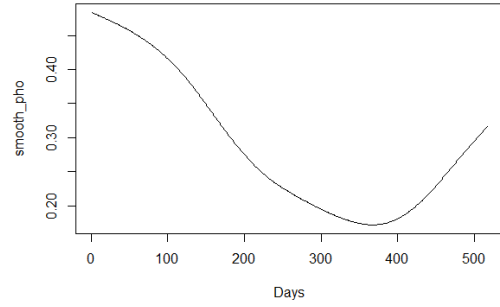
We make plots of lower tail coefficient & upper tail coefficient vs threshold values for select value of parameter vectors: $\Theta_1 = [0.14285576 \ 0.05672977 \ 0.45728173]^T$, $\Theta_2 = [0.3488898 \ 0.3850638 \ 0.2261966]^T$, $\Theta_3 = [0.4000281 \ 0.1910818 \ 0.3489596]^T$, $\Theta_4 = [0.3546800 \ 0.3597338 \ 0.1741803]^T$.

The plots obtained show that near $t \rightarrow 0$ for lower tail coefficient $\chi_l \neq 0$ showing lower tail dependence in the dataset. Similarly for $t \rightarrow 1$ for upper tail coefficient $\chi_u \neq 0$ showing upper tail dependence in the dataset.

From the values of $\delta_{l/u}$ of the smooth curve we obtain the following plots of lower/upper tail coefficients η_l, η_u .

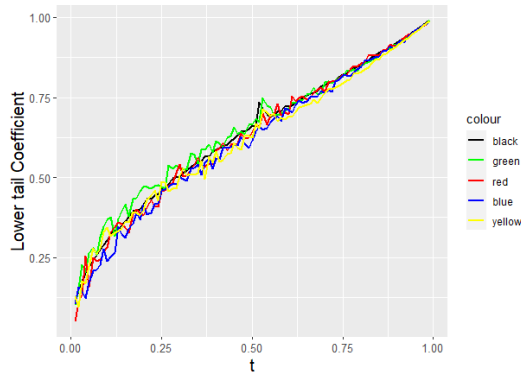


(a) ρ vs days from 21-01-2020 to 10-04-2022

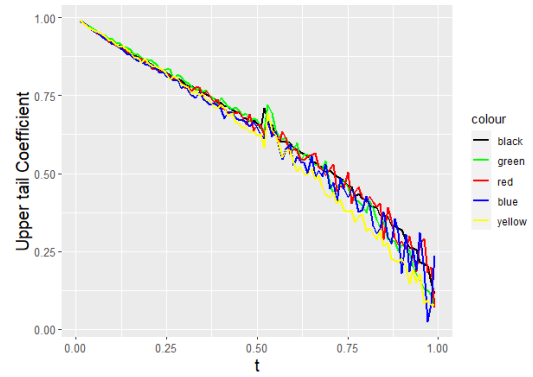


(b) ρ (smooth curve) vs days from 21-01-2020 to 10-04-2022

Figure 11: Time varying plots of parameter ρ

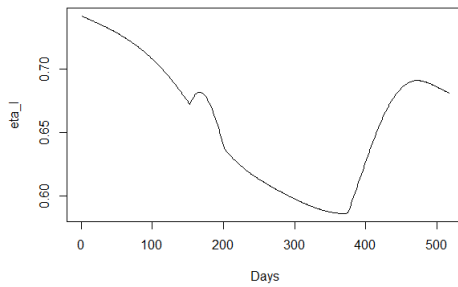


(a) χ_l vs tolerance value

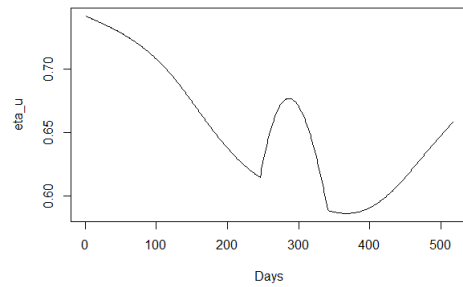


(b) χ_u vs tolerance value

Figure 12: Tail coefficients vs tolerance value for parameters $\Theta_1, \dots, \Theta_4$



(a) η_l vs days from 21-01-2020 to 10-04-2022



(b) η_u vs days from 21-01-2020 to 10-04-2022

Figure 13: Time varying plots of $\eta_{l/u}$

CONCLUSION

In our research, we introduced a novel method for analyzing the behavior of aviation stock prices during the COVID-19 pandemic. This methodology combines two key components: the use of copulas for modeling dependence and EVGAM (Extreme Value Generalized Additive Models) for modeling the marginal distributions as well as Copula model from Gong & Heuser’s research paper.

This approach offers a comprehensive perspective on the behavior of stock prices during a time of extreme economic uncertainty.

The real data analysis in our study is focused on the stock prices of two major airlines.

One of the key applications of our research is portfolio optimization. Understanding the dependence structure between the stock prices of these two airlines during the pandemic enables investors and portfolio managers to make more informed decisions. This is especially important during times of economic upheaval, where risk management and optimized asset allocation become crucial.

The emphasis on the COVID-19 pandemic period is significant. It allows us to delve into the unique and extreme characteristics of this period, providing insights into how extreme events impact stock prices and risk. Our results indicating ”extremal dependence at sub-asymptotic levels for both the tails” suggest that the relationship between these aviation stocks during the pandemic was highly dependent, particularly in extreme scenarios. This information is invaluable for those looking to understand, manage, and optimize portfolios in uncertain times.

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to Professor Arnab Hazra for his invaluable guidance and support throughout my project. His expertise, encouragement, and insights greatly contributed to my work. I am truly thankful for his mentorship and assistance.

BIBLIOGRAPHY

- [1]Gong, Y. and Huser, R., Asymmetric tail dependence modeling, with application to cryptocurrency market data, *Annals of Applied Statistics*, Volume 16, Issue 3, 1822–1847
- [2]Coles, Stuart, Classical Extreme Value Theory and Models, An Introduction to Statistical Modeling of Extreme Values, Springer London, 2001
- [3]Bose, Shiladitya and Hazra, Arnab, Estimating Changepoints in Extremal Dependence, Applied to Aviation Stock Prices During COVID-19 Pandemic, arXiv e-prints, 2023 Aug'
- [4]Wikipedia Contributors, Copula (probability theory), Wikipedia, The Free Encyclopedia
- [5]Wikipedia contributors, Stationary process, Wikipedia, The Free Encyclopedia
- [6]Benjamin D. Youngman, evgam: An R package for Generalized Additive Extreme Value Models, arXiv, 2020